

## DAY-6

### Wireless Standard & Configuration : Wireless Setup, Channel Selection & Avoiding Interference, Wireless Security, MAC Filtering & Access Control Lists

**Q-1 Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.**

**[A]**

- To understand basic wireless configuration, channel selection, wireless security, MAC filtering, and router ACLs. The practical covers setting up a Wi-Fi network, reducing interference, securing the wireless network, controlling device access, and restricting access to a specific destination.

#### ▪ **Network Setup**

| Setting         | Value             |
|-----------------|-------------------|
| Router          | Home Wi-Fi Router |
| SSID            | College_WiFi      |
| Wireless Band   | 2.4 GHz           |
| Default Channel | Auto              |
| Security        | WPA2-Personal     |
| Password        | College@WiFi2026  |

#### ▪ **Procedure**

1. Power on the Wi-Fi router.
2. Connect your laptop or phone to the router.
3. Open the router's administration page.
4. Go to **Wireless/Wi-Fi Settings**.
5. Note the SSID and current channel.

**Q-2 Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks. Hint: Look for overlapping channels and pick one with the least overlap.**

**[A]**

- **Use Wi-Fi Analyzer**

- Install a Wi-Fi analyzer application on your phone and scan nearby networks.

| Channel | Nearby Networks | Interference |
|---------|-----------------|--------------|
| 1       | 6               | Medium       |
| 6       | 2               | Low          |
| 11      | 7               | High         |

- **Procedure**

1. Open the Wi-Fi analyzer.
2. Scan nearby wireless networks.
3. Check which channels are heavily occupied.
4. Select the channel with the least interference.
5. Configure that channel manually on the router.
6. Reconnect the wireless device.
7. Perform the scan again.

**Q-3 Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.**

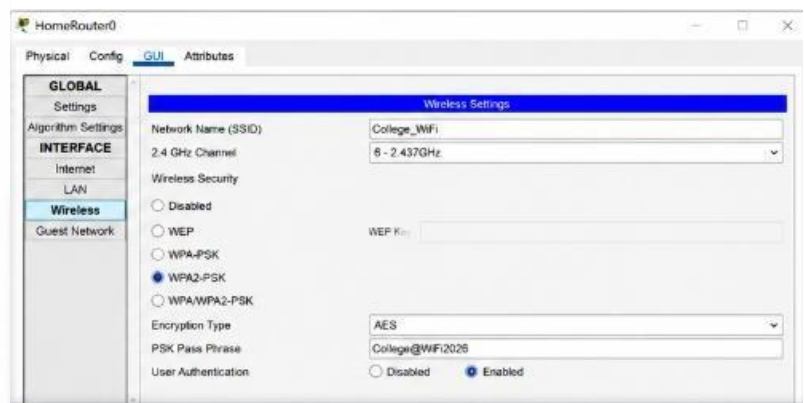
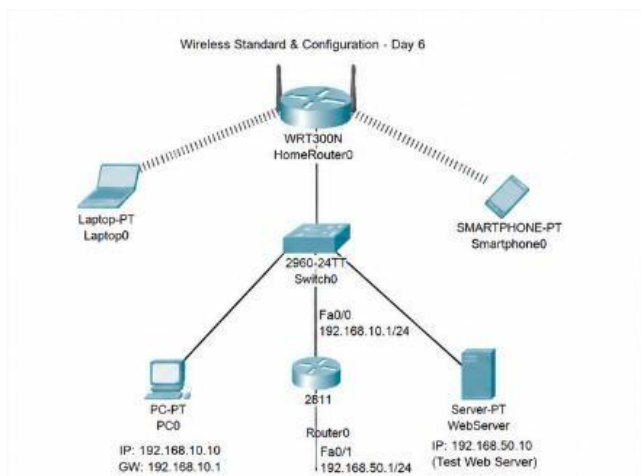
**[A]**

- Recommended configuration:
  - Security Mode: WPA2-Personal ○ Encryption: AES ○ SSID: College\_WiFi
  - Password: College@WiFi2026
  
- If your router supports WPA3:
  - Security Mode: WPA3-Personal ○ Use a strong password containing uppercase/lowercase letters, numbers, and symbols.
  
- Example:
  - College@WiFi#2026!

### ➤ Test the Security

On the laptop or phone:

1. Forget the existing Wi-Fi network.
2. Search for College\_WiFi.
3. Select the network.
4. Enter the new password.
5. Connect.





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Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router#show access-lists
Extended IP access list 100
 10 deny tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www (18 matches)
 20 deny tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq 443 (11 matches)
 30 permit ip any any (25 matches)
Router#
  
```

**Q-4 Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result. Hint: You can find your device's MAC address in the network settings.**

**[A]**

- MAC filtering allows the router to permit or deny devices based on their **MAC address**.

- **Find the Device MAC Address**

- On Windows: ipconfig /all
- Look for:  
Physical Address
- Example:  
A4-B1-C1-22-33-44

- **Test with an Unauthorized Device**

| Device | MAC Address       | Status  |
|--------|-------------------|---------|
| Laptop | A4:B1:C1:22:33:44 | Allowed |
| Phone  | 78:25:AD:55:66:77 | Denied  |

**Q-5 Create an Access Control List (ACL) on your router to block access to a specific website (for example, [www.youtube.com](http://www.youtube.com)) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.**

**[A]**

- An ACL can be used to control traffic based on IP addresses, protocols, ports, and other packet characteristics.
- For a simple **Cisco Packet Tracer lab**, it is better to block a specific **destination IP address** representing the website/server.

▪ **Important Practical Note**

- A normal router ACL does **not reliably block a website by its name**, such as [www.youtube.com](http://www.youtube.com), because modern websites can use multiple changing IP addresses and HTTPS. For a classroom demonstration, use a known test-server IP or a controlled web server.

▪ **Example Extended ACL**

- Enter:  
Router> enable  
Router# configure terminal
- Create the ACL:  
Router(config)# access-list 100 deny tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq 80  
Router(config)# access-list 100 deny tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq 443  
Router(config)# access-list 100 permit ip any any
- Apply it to the interface:  
Router(config)# interface gigabitEthernet 0/0  
Router(config-if)# ip access-group 100 in  
Router(config-if)# exit

▪ **Practical Observation Table**

| Task              | Configuration                  | Observation                   |
|-------------------|--------------------------------|-------------------------------|
| Wireless Setup    | SSID = College_WiFi            | Device connected successfully |
| Channel Selection | Channel changed from Auto to 6 | Lower interference observed   |
| Wireless Security | WPA2/WPA3 enabled              | Correct password required     |
| MAC Filtering     | Authorized MAC added           | Authorized device connected   |
| ACL               | Test server blocked            | Web access was denied         |